

Checkpoint 3 — Identify the Posterior for a Conjugate Pair

INFO 521 — Machine Learning Foundations

i Course-schedule placement

Week 4 · with Module 4 — Bayesian Inference (conjugacy) · gates Project 1.3.

Gates: Project 1.3 (Bayesian) **Format:** in-class, closed-book, ~15 min. Graded Satisfactory / Not-Yet. **Retakeable.**

This checkpoint tests the study-guide outcome “*given a conjugate prior–likelihood pair, identify the form of the posterior.*” It uses Beta–Binomial as the clean, time-boxed instance. The project applies the **same principle** to the Gaussian–Gaussian pair — make that connection explicit when you hand it back.

Prompt (Version A)

A coin has unknown heads probability r . You place a prior $r \sim \text{Beta}(\alpha, \beta)$ and observe y heads in N tosses (Binomial likelihood). Identify the posterior distribution of r — name the family and give its parameters. State in one sentence what makes this pair *conjugate*.

Prompt (Version B — retake)

You place a Gaussian prior $w \sim \mathcal{N}(0, \tau^2)$ on a single weight in a Gaussian-noise linear model. Without full derivation, name the family of the posterior $p(w \mid \text{data})$ and explain why conjugacy guarantees it stays in that family.

Satisfactory criteria

- Names the correct posterior family.
- (A) Gives correct updated parameters.
- States conjugacy = prior and posterior share a family (posterior stays analytically tractable).

Answer key

A: Posterior is Beta $(\alpha + y, \beta + N - y)$. Conjugate because the Beta prior and Binomial likelihood combine to yield another Beta.

B: Posterior is **Gaussian**. A Gaussian prior with a Gaussian-noise likelihood yields a Gaussian posterior, so it remains closed-form — the exact property Project 1.3 relies on and Project 2 deliberately breaks.